VIT

Vellore Institute of Technology

Reg. No. : 22BAI1266

Final Assessment Test (FAT) - May 2024

Programme	B.Tech.	Semester	WINTER SEMESTER 2023 - 24
Course Title	MACHINE LEARNING	Course Code	BCSE209L
Faculty Name	Prof. Syed Ibrahim S P	Slot	C1+TC1
		Class Nbr	CH2023240501696
Time	3 Hours	Max. Marks	100

General Instructions:

- Write only Register Number in the Question Paper where space is provided (right-side at the top) & do not write any other details.

Section - I

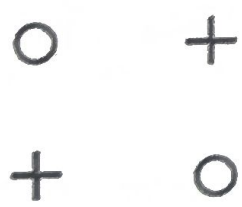
Answer all questions (5 X 10 Marks = 50 Marks)

01. A food delivery company which performed well for 2 years is not able to perform well in recent months as they are not able to deliver food on time. [10]

As a result, their customers are unhappy. So, the company has requested you to help them in improving their business.

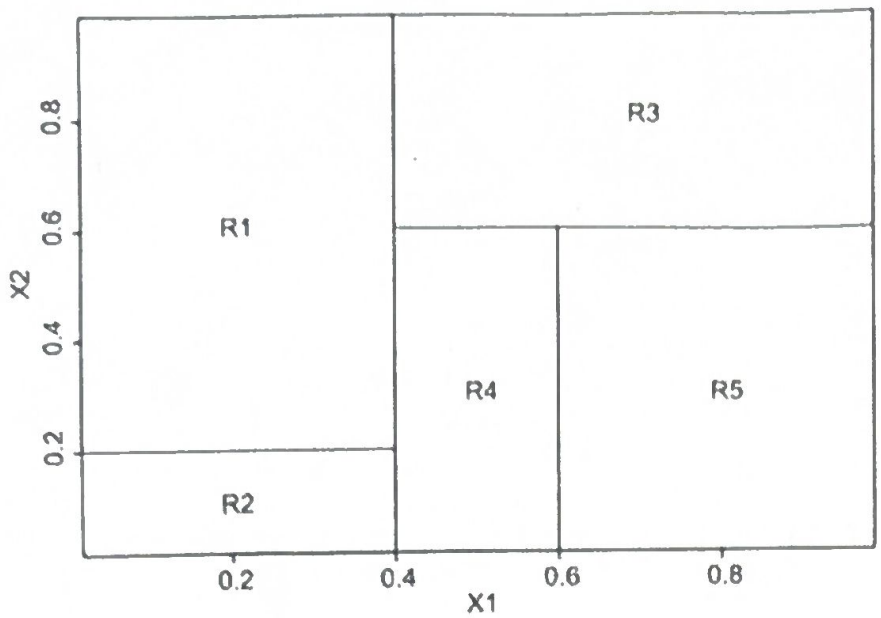
As a machine learning expert, provide them with any 5 major solutions along with corresponding ML algorithms that could be applied for each of the scenario to help them recover from this situation?

02. i) Consider the following dataset represented as data points: [10]



Discuss how classifiers like Logistic regression and 3-NN can classify the train data? Is it possible for the above list of classifiers to achieve zero training error on this data set. Justify your answer. (5 Marks)

ii) Draw a decision tree having zero training error for the given scenario having 5 regions R1, R2, R3, R4, R5 (which are to be the leaf nodes in the DT) using the appropriate attribute ranges. (5 marks)



03. [10]

No	Petal Length	Petal Width	class
1.	4.8	1.8	Iris-virginica
2	3.7	1.2	Iris-setosa
3	4.5	1.4	Iris-setosa
4	3.9	1.2	Iris-setosa
5	4.1	1.3	Iris-setosa
6	3.6	1.6	Iris-setosa
7	5.5	1.8	Iris-virginica
8	3.5	1.3	Iris-setosa

i) Analyze the given dataset and check whether any ML classifier model applied on it can produce consistent level of accuracy. Justify your answer. (3 marks)

ii) Recommend a suitable mechanism to get the best possible accuracy for the model. Show the step-by-step process of applying the recommended mechanism to solve the problem in the above given data set. (7 marks)

04. A telecom company is willing to share the following details with you, Data were provided for 1,00,000 customers with last 6 months of service history [10]

– Having churners and non-churners
– Type and price of current handset
– Date of last handset change/upgrade
– Total revenue
– Call behaviour statistics (type, number, duration, totals, etc.)
– Demographic information

based on the above information, answer the following question.

Assume that your ML algorithm has achieved 70 % accuracy even after trying with various hyper parameter tuning, but you are required to achieve a minimum 80% accuracy.

i) Recommend possible mechanisms that can improvise on the accuracy to 80%. (3 marks)

ii) For the above scenario show the working of any one recommended approach with proper illustration. (7 marks)

05. i) Justify the need for generating optimal set of actions for a given goal based agent. (3 marks) [10]

ii) Illustrate with a suitable example for at least 2 iterations an off-policy based approach to achieve the optimal path with max rewards (7 marks)

Section - II

Answer all questions (2 X 15 Marks = 30 Marks)

06. A company has 6 equipments and each fitted with 3 sensors. They have collected equipment failure details based on data referencing observations of past machine runs and failures as shown in the table below. Use this information to answer below questions. [15]

Table: Machine Failure Status data

Machine Id	Sensor1	Sensor 2	...	Sensor 300	Machine Failure
M1	0.310	.399	...	0.333	No
...	...	...	...	...	...
M500	0.748	0.329	...	0.938	Yes

Assume that there are equal number of class labels in the above data and are linearly separable.

i) Draw a Neural Network architecture with corresponding activation function to classify the data. Also, mention clearly the steps on how the NN you suggested learns from its erroneous predictions. (7.5 marks)

ii) Provide detailed description of a suitable classifier that can provide a separator which can maximize the distance of the closest datapoint from the decision line / hyperplane. (7.5 marks)

07. a) Consider the table Machine failure status data. You are asked to perform unsupervised learning on this data set. In this context, identify the problem posed by the dataset. Explain briefly a possible solution to overcome the problem posed by the data.(3 marks) [15]

Table: Machine Failure Status data

Machine Id	Sensor1	Sensor 2	...	Sensor 300	Machine Failure
M1	0.310	.399	...	0.333	No
...	...	...	...	...	...
M500	0.748	0.329	...	0.938	Yes

b) Assume that you have been assigned a task of grouping similar customers from the table computer purchase. For this given task, identify the suitable grouping algorithm and justify your choice. (3 marks)

c) Consider the customer samples ID1, ID4, ID12 as 3 initial points for grouping. Apply the algorithm identified in (b) and show step by step procedure to obtain the groups formed upto 2 iterations.(9 marks)

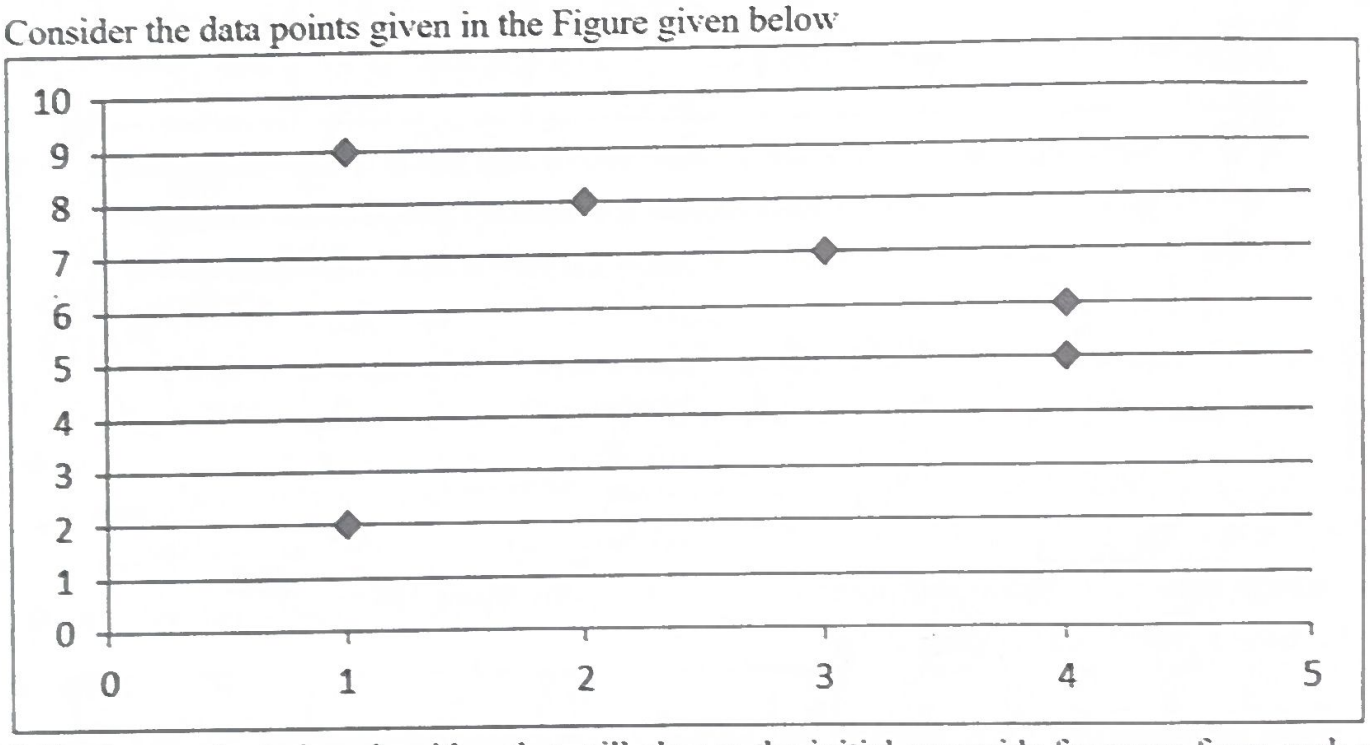
Table: Computer purchase

ID	AGE	INCOME	STUDENT	CREDIT	BUYS
1	< 31	high	no	bad	no
2	< 31	high	no	good	no
3	31 – 40	high	no	bad	yes
4	> 40	med	no	bad	yes
5	> 40	low	yes	bad	yes
6	> 40	low	yes	good	no
7	31 – 40	low	yes	good	yes
8	< 31	med	no	bad	no
9	< 31	low	yes	good	yes
10	> 40	med	yes	bad	yes
11	< 31	med	yes	good	yes
12	31 – 40	med	no	good	yes
13	31 – 40	high	yes	bad	yes
14	> 40	med	no	good	no

Section - III

Answer all questions (1 X 20 Marks = 20 Marks)

08. Consider the data points given in the Figure given below [20]



i) Perform a clustering algorithm that will choose the initial centroids far away from each other. (10 marks)

ii) Use Self organizing map with the above data and compare the SOM output with that of clustering output and provide the inference. (10 marks)

Note - Do the above computations for 1 iteration.